NutriScan

UI/UX Design & Frontend Challenge

Career Fair • 2-Day Challenge

Duration	2 Days
Difficulty	Beginner → Intermediate
AI Allowed	Yes — use any AI tool, design assistant, or code generator you like
Deliverable	Wireframes for desktop + mobile, then a fully built desktop web app
Backend	Not required — use mock data and browser/local storage only

Overview

NutriScan is a calorie tracking app. Users can log their meals throughout the day, see their nutritional intake, and track progress toward a daily calorie goal.

This challenge has two parts. First, you will think and design, producing wireframes for both desktop and mobile views. Then you will build your desktop wireframe into a real, working web application. No backend is needed; all data lives in the browser.

This challenge is about product thinking as much as coding. Judges will evaluate how well your final app reflects your original design intent, and how thoughtful your UI and UX decisions are.

Part 1 - Design

Before writing a single line of code, you must produce wireframes. Wireframes are low-to-medium fidelity blueprints of your app they show layout, structure, and user flow without requiring final visual polish.

What to Design

You must produce wireframes for two views of the same app:

Desktop View (min. 1280px wide)

- Dashboard / home screen — daily calorie summary, progress toward goal, today's meal log
- Add meal screen or panel — search or select a food, enter quantity, confirm
- Meal history view — a list or timeline of logged meals with calorie breakdown

Mobile View (375px wide — iPhone standard)

- Same three screens adapted for a small screen. Navigation changes, layout stacks vertically
- Consider thumb reach: key actions should be reachable at the bottom of the screen
- **You will NOT build the mobile version; this is a design-only exercise for mobile**

Wireframe Rules

- Use any tool you like Figma, Draw.io, Balsamiq etc...
- Label every screen and every key UI element
- Show the user flow how does a user get from the dashboard to adding a meal?
- Submit wireframes as image files or a shareable link alongside your code

What Makes a Good Wireframe

- Clear visual hierarchy the most important information is the most prominent
- Consistent layout patterns across screens
- Realistic content uses real food names, real calorie numbers, not 'Lorem Ipsum'
- Mobile version shows real adaptation, not just a shrunk desktop layout

Part 2 - Build the Desktop Web App

Any Language and frameworks are allowed (React, Vue, plain JS, Angular, etc...).

There is no backend. Store all data in local Storage or in-memory JavaScript state. The app must work by simply opening an HTML file or running a local dev server.

Core Features the App Must Have

- A daily calorie goal that the user can set
- A meal log where users can add food entries with a name, calorie count, and meal type (breakfast, lunch, dinner, snack)
- A running total of calories consumed today vs the daily goal
- The ability to delete a logged meal entry
- Data persists on page refresh using localStorage

Challenge Levels

Each level builds on the previous one. Level 1 covers the wireframe and core app, although Levels 2 and 3 add depth to the build.

LEVEL 1 - Design + Core Tracker

Requirements

- Submit wireframes for all 3 screens in both desktop and mobile views
- Build the desktop dashboard showing today's calorie total vs goal
- Implement the add meal form — name, calories, meal type
- Display logged meals in a list with a delete option
- Persist data in localStorage so it survives a page refresh

Technical Hints

- *Start with the wireframe do not code until your layout is sketched out*
- *Use a JavaScript array as your data store, then sync it to localStorage on every change*
- *JSON.stringify() to save, JSON.parse() to load from localStorage*
- *A simple flexbox or CSS grid layout is enough for the dashboard*
- *Meal types can be a hardcoded dropdown: Breakfast, Lunch, Dinner, Snack*

LEVEL 2 - Richer Experience

Requirements

- Add a visual progress indicator a progress bar or ring showing % of daily goal consumed

Technical Hints

- *A CSS progress bar is just a div with a width set to a percentage — keep it simple*
- *Group meal log entries by type using Array.filter() or Array.reduce()*

• Show a per-meal-type calorie breakdown (how many calories from breakfast vs lunch, etc.) • Allow the user to edit a logged meal entry after adding it • Add basic input validation — calories must be a positive number, name must not be empty • Show a warning or visual cue when the user exceeds their daily goal	• *For editing: toggle between a display view and an input view for each meal row* • *Validation: check inputs before pushing to the array, show an inline error message* • *Over-goal warning: compare total calories to goal and conditionally apply a CSS class*

LEVEL 3 - Polish and Delight

Requirements	Technical Hints
• Your final app must closely reflect your original desktop wireframe • The UI must be visually consistent — a clear color palette, typography, and spacing system	• *Store daily logs by date key in localStorage — e.g. { '2025-04-15': [...meals] }*

• Add a meal history view showing logs from previous days • Implement a simple food library — a pre-built list of common foods with preset calories the user can pick from • Bonus: add a simple chart (bar or line) showing calorie intake across the past 7 days	• *A food library is just a hardcoded array of { name, calories } objects shown in a dropdown or search list* • *For the chart bonus: Chart.js is a lightweight library that works with a single script tag — no install needed* • *Use CSS custom properties (variables) for colors so your theme stays consistent across the whole app* • *Run your app at 1280px wide and check it matches your wireframe before the demo*

Judging Criteria

Maximum score: 100 points. Judges will review your wireframes and then demo your live app.

Criteria	What We Look For	Points
Wireframe Quality	Clear, labeled, realistic desktop and mobile both present	**25 pts**

Design Consistency	App visually matches the wireframe, cohesive color and layout	**20 pts**
Core Functionality	Meal logging, calorie totals, delete, and localStorage all work	**20 pts**
Level Completion	How many levels were fully and correctly completed	**15 pts**
UX Thoughtfulness	The app is intuitive, handles errors, gives clear feedback	**15 pts**
Code Quality	Clean, readable, well-structured code	**5 pts**

- Extra 15 points for having it deployed on GitHub pages.

Tips for Success

- Wireframe first, always. Students who skip the design step and go straight to code usually end up with a messy, inconsistent UI.
- Your mobile wireframe does not need to be built — but it does show judges you understand responsive design thinking.
- Use realistic data in your demo: real food names, real calorie values, and a realistic daily goal.
- A simple, polished Level 1 app beats a half-finished Level 3 app every time.
- Use AI freely for code generation, design ideas, or debugging. The final product is what counts.